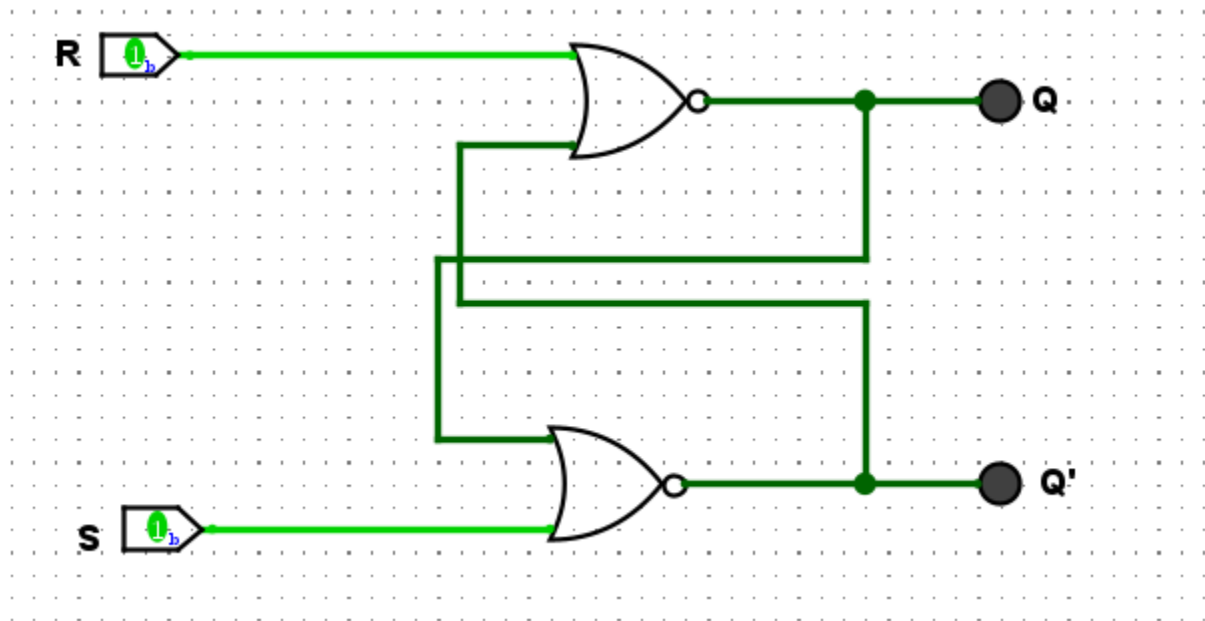


6.

set	reset	Q	Q'
1	0	1	0
1	1	0	0
0	1	0	1
1	1	0	0

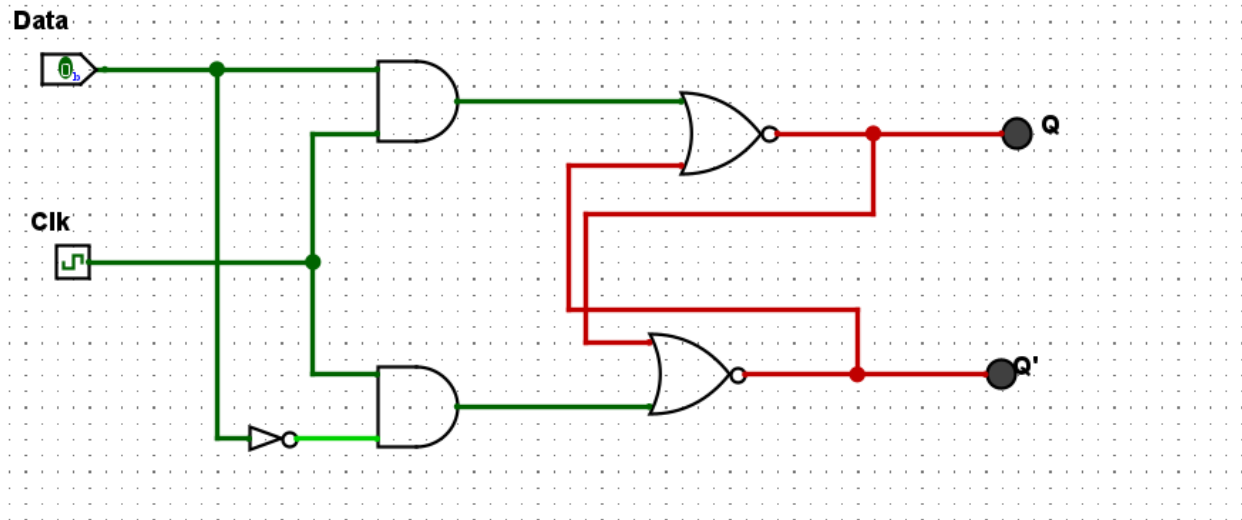


7. One input at 1 set or reset the state, this lets the circuit store a bit.

8. Both inputs at 1 cause an invalid state, Q and Q' are no longer opposites, which breaks logic.

10.

Clock	Pin	Q	Q'
0	0	0	0
0	1	0	0
1	1	0	1
1	0	1	0

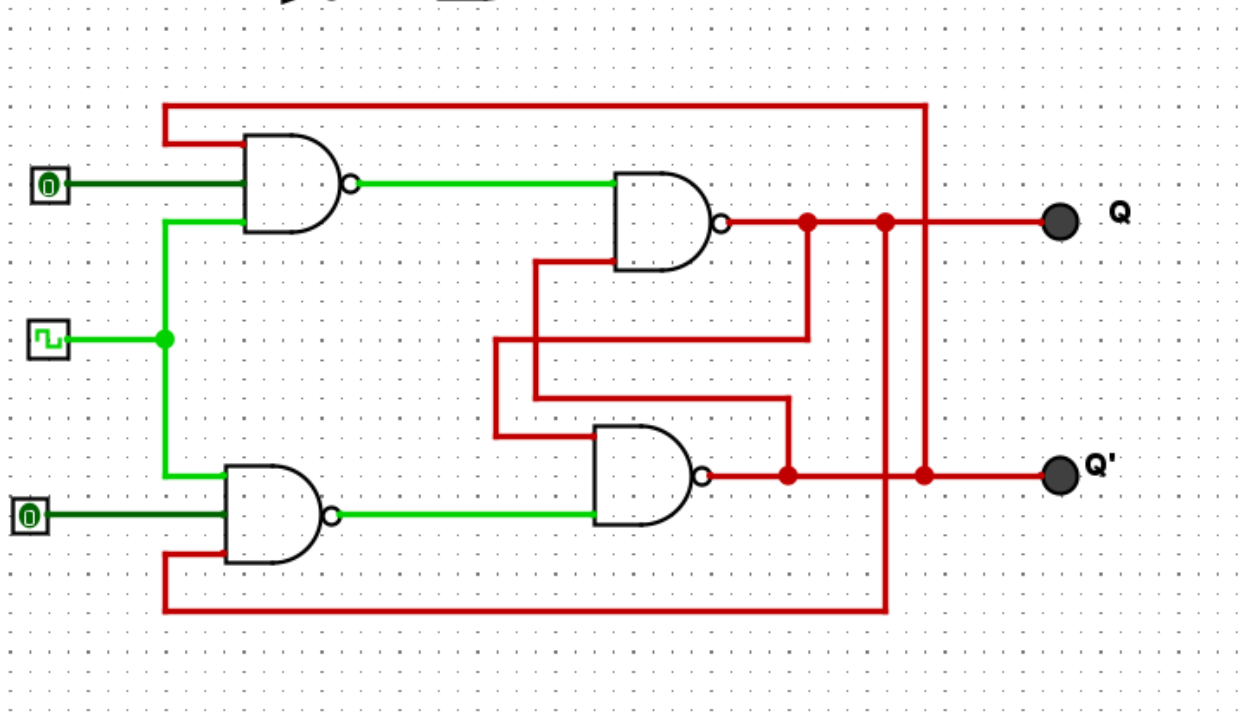


11. It stores the value of D only when the clock is high. This makes it ideal for syncing data flow in digital systems.

12. The clock controls when the flip-flop updates. Q and Q' only change on a clock pulse, ensuring predictable timing.

13. Unlike the R-S flip-flop, the D Flip-Flop avoids invalid states and simplifies input logic—just one data line and a clock.

15.



16. JK as D Flip-Flop, set $J = D$ and $K = \text{NOT } D$. This makes it act like a D Flip-Flop.

17. JK as T Flip-Flop, set $J = 1$ and $K = 1$. This makes it toggle the output on each clock pulse.

22.

Ox	Input Binary	Output Binary
0	0000	0000
1	0001	0001
2	0010	0010
3	0011	0011
5	0101	0101
A	1010	1010
B	1011	1011
C	1100	1100
D	1101	1101
E	1110	1110
F	1111	1111

